Physics P1: Physical Reasoning and Estimation

MAIS Physics & Astronomy Club (PAC)

Before solving differential equations or memorizing long formulas, physicists ask simpler questions: **Do the units match? What happens in extreme cases? Roughly how big should the answer be?** These three habits catch algebraic mistakes immediately and often reveal the form of a physical law before doing a single integral. Problems are marked from **[1]** (warm-up) to **[3]** (challenge).

1. Units and Dimensional Analysis

Every physical quantity in mechanics can be built from three base dimensions: length [**L**], mass [**M**], and time [**T**]. Two quantities may be added, subtracted, or equated only if their dimensions match, and the argument of any transcendental function (such as sin **x**, $\mathbf{e^x}$, or ln **x**) must be dimensionless.

Example 1. A simple pendulum consists of a small bob of mass **m** hanging on a light string of length ℓ in a uniform gravitational field **g**. How does the period **P** of small oscillations depend on **m**, ℓ, and **g**?

Solution. Write $\mathbf{P} = \mathbf{C}\,\mathbf{m}^{\mathbf{a}}\,\ell^{\mathbf{b}}\,\mathbf{g}^{\mathbf{c}}$, where **C** is a dimensionless constant. Comparing dimensions gives

$$[\mathbf{T}] = [\mathbf{M}]^{\mathbf{a}}\,[\mathbf{L}]^{\mathbf{b}}\,([\mathbf{L}][\mathbf{T}]^{-2})^{\mathbf{c}} = [\mathbf{M}]^{\mathbf{a}}\,[\mathbf{L}]^{\mathbf{b}+\mathbf{c}}\,[\mathbf{T}]^{-2\mathbf{c}}. \tag{1}$$

Matching exponents of [**M**], [**T**], and [**L**] yields $\mathbf{a} = 0$, $-2\mathbf{c} = 1 \Rightarrow \mathbf{c} = -1/2$, and $\mathbf{b} + \mathbf{c} = 0 \Rightarrow \mathbf{b} = 1/2$. Therefore

$$\mathbf{P} = \mathbf{C}\,\sqrt{(\ell)/\,(\mathbf{g})}. \tag{2}$$

Newton's laws are only needed to fix the numerical prefactor $\mathbf{C} = 2\pi$. Even without solving the equation of motion, dimensional analysis already tells us that doubling the mass has no effect, while quadrupling the length doubles the period.

Remark. Why did we restrict to **small** oscillations? If the release angle θ_0 is large, θ_0 is itself a dimensionless parameter, so **C** becomes a function $\mathbf{f}(\theta_0) = 2\pi(1 + \theta_0^{\,2}/16 + \cdots)$. Always check whether hidden dimensionless ratios exist!

[1] Problem 1. Ocean ripples and waves carry energy across water of density ρ under gravitational acceleration $\mathbf{g}$.

(a) For waves of wavelength λ on very deep water (depth $\mathbf{h} \gg \lambda$), the wave speed $\mathbf{v}$ can depend only on λ, $\mathbf{g}$, and ρ. Use dimensional analysis to find how $\mathbf{v}$ scales with λ, $\mathbf{g}$, and ρ.

(b) For tsunami waves in an ocean whose depth $\mathbf{h}$ is much smaller than the wavelength ($\mathbf{h} \ll \lambda$), experiments show that $\mathbf{v}$ is independent of λ. Find how $\mathbf{v}$ depends on $\mathbf{h}$ and $\mathbf{g}$, and estimate the speed (in km/h) of a tsunami over a 4.0 km deep ocean basin.

2. Limiting Cases and Symmetry

Whenever you derive an algebraic expression, test it against cases where you already know what must happen physically: setting an angle to 0 or 90°, letting one mass become much heavier than another ($\mathbf{M} \gg \mathbf{m}$), or swapping two identical labels.

Example 2. Two blocks of masses $\mathbf{m}_1$ and $\mathbf{m}_2$ hang on opposite sides of a massless, frictionless pulley (an Atwood machine). Suppose a student writes down $\mathbf{a} = (\mathbf{m}_1 + \mathbf{m}_2)/(\mathbf{m}_1 - \mathbf{m}_2)\,\mathbf{g}$ for the downward acceleration of $\mathbf{m}_1$. How do we see immediately that this is wrong?

Solution. If $\mathbf{m}_1 = \mathbf{m}_2$, the system is balanced and must have $\mathbf{a} = 0$, whereas the student's formula diverges to infinity! The correct result,

$$\mathbf{a} = (\mathbf{m}_1 - \mathbf{m}_2)/(\mathbf{m}_1 + \mathbf{m}_2)\,\mathbf{g}, \tag{3}$$

vanishes when $\mathbf{m}_1 = \mathbf{m}_2$, approaches $+\mathbf{g}$ (free fall) when $\mathbf{m}_1 \gg \mathbf{m}_2$, and changes sign when $\mathbf{m}_1$ and $\mathbf{m}_2$ are interchanged.

[2] Problem 2. A small block of mass $\mathbf{m}$ is placed on a frictionless wedge of mass $\mathbf{M}$ and inclination angle θ, which itself rests on a frictionless horizontal table. Five candidates are proposed for the magnitude $\mathbf{A}$ of the wedge's horizontal acceleration:

(i) $(\mathbf{mg}\cos\theta)/(\mathbf{M}+\mathbf{m})$, (ii) $(\mathbf{Mg}\sin\theta\cos\theta)/(\mathbf{M}+\mathbf{m}\sin^2\theta)$, (iii) $(\mathbf{mg}\sin\theta\cos\theta)/(\mathbf{M}+\mathbf{m}\sin^2\theta)$, (iv) $(\mathbf{mg}\tan\theta)/(\mathbf{M}+\mathbf{m})$, (v) $(\mathbf{mg}\sin\theta\cos\theta)/(\mathbf{m}+\mathbf{M}\sin^2\theta)$.

(a) By examining limiting cases such as $\theta \to 0$, $\theta \to \pi/2$, $\mathbf{M} \to \infty$, and $\mathbf{M} \gg \mathbf{m}$ at small θ (without solving Newton's equations from scratch), eliminate four of the five expressions and identify the only physically consistent candidate.

(b) Verify the surviving formula directly by writing down Newton's second law for the block and the wedge.

3. Order-of-Magnitude Estimation

An order-of-magnitude (or Fermi) estimate breaks an unfamiliar quantity into pieces you know within a factor of 2 or 3. Overestimates and underestimates tend to partially cancel when multiplied together, frequently landing remarkably close to the exact value.

[2] Problem 3. Sea-level atmospheric pressure is $\mathbf{P}_0 \approx 1.01 \times 10^5$ Pa, Earth's radius is $\mathbf{R}_\oplus \approx 6.37 \times 10^6$ m, and $\mathbf{g} \approx 9.8$ m/s^2.

(a) Explaining why the atmosphere's thickness (≈ 10 km) being small compared to $\mathbf{R}_\oplus$ simplifies the calculation, estimate the total mass $\mathbf{M}_{\text{atm}}$ of Earth's atmosphere.

(b) Ulaanbaatar sits at an elevation of roughly 1350 m above sea level, where typical atmospheric pressure is about 8.6×10^4 Pa. Using the same idealized vertical-column model, approximately what fraction of a sea-level air column's mass lies **below** the altitude of Ulaanbaatar? (The global fraction below 1350 m also depends on topography.)

[3] Problem 4 (Collapsing clouds and stars). Suppose all internal pressure inside a spherical cloud or star of uniform initial density ρ_0 suddenly vanished, so that it collapsed inward solely under its own gravity (with Newton's constant $\mathbf{G} = 6.67 \times 10^{-11}$ N·m^2/kg^2).

(a) Using dimensional analysis, find how the collapse time $\mathbf{t}_{\text{ff}}$ depends on $\mathbf{G}$, ρ_0, and the initial radius $\mathbf{R}_0$. Explain physically why $\mathbf{R}_0$ drops out when ρ_0 is held fixed.

(b) Estimate $\mathbf{t}_{\text{ff}}$ in minutes for the Sun (mass $\mathbf{M}_\odot = 1.99 \times 10^{30}$ kg, radius $\mathbf{R}_\odot = 6.96 \times 10^8$ m). Why has the real Sun shone steadily for 4.6 billion years instead of collapsing on this timescale?